

AEC 8123: Analysis of Agricultural Markets

Dr. Ayoung Kim

Associate Professor

Agricultural Economics, Mississippi State University

Syllabus: Spring 2027

Course Info

3 Credits, In-person Lecture or Online (for Campus 1 and Campus 5)[†]

Class Meetings:

Tuesday and Thursday, 9:00 - 10:15 AM CST

Contact Info

Email: a.kim@msstate.edu

Office Hours: by appointment (in person or virtually, depending on the delivery format)

Students are encouraged to email to schedule a meeting.

Virtual Meeting Room in Webex: <https://msstate.webex.com/meet/ak1896>

[†]Regardless of the course delivery format, all exams will be conducted in person.

* Disclaimer: This syllabus is subject to change. Students will be notified of any minor revisions, which will be posted on Canvas. (last updated on 09/21/2026)

Course Description

(From the course catalog) Three-hour lecture. This course integrates economic theory with econometric methods, focusing on models of consumer demand and agricultural supply under perfect and imperfect competition settings. It covers modern estimation techniques, identification strategies, and (approaches to causal inference).

The underlying goal of this course is to familiarize the student with the application of the scientific method in agricultural economics. We will concentrate on four major empirical frameworks: (1) consumer demand, (2) agricultural supply, (3) supply and demand, and (4) market structure and imperfect competition.

Prerequisites

The student must have basic knowledge of calculus, economics, and computer skills. The course will frequently use algebra, calculus, and R programming languages. A basic understanding of statistics and econometrics is also required. Two core tools will be used in this class: mathematical optimization and regression analysis.

Course Website

The course will utilize Canvas (canvas.msstate.edu) for all class-related announcements, including important schedule updates. Course materials, including lecture notes and/or videos, readings, quizzes, answer keys, discussions, and other instructional materials, will be available through Canvas. Course materials and announcements will be posted on Canvas according to the course schedule.

Technology Requirements

You will need a computer with access to high-speed Internet. For this course, you should have access to the following technologies and software:

- A modern web browser (e.g., Chrome, Firefox, or an equivalent)
- Cisco Webex or another university-supported platform may be used for scheduled virtual meetings in online sections (camera and microphone recommended).
- Adobe Acrobat Reader (or equivalent PDF viewer) for viewing and annotating course materials
- A PDF scanner app (not required but useful), such as CamScanner, Adobe Scan, or any similar app you prefer for creating PDF files from handwritten work

In addition, students may find it helpful to use note-taking or organization tools (e.g., OneNote, Notability, or GoodNotes) to manage lecture notes and assignments. You are welcome to use any apps or tools that work best for your workflow, and I encourage you to share helpful tools or tips with your classmates throughout the semester.

Use of Generative AI

Generative AI tools (e.g., ChatGPT or similar technologies) may be used as a learning aid for reviewing concepts, checking understanding, or exploring alternative explanations. However, all submitted work must reflect the student's own reasoning and understanding. Direct copying of AI-generated content without substantial modification or understanding is not permitted. Students should be aware that assessments in this course, particularly exams and other individual assessments, are designed to evaluate independent analytical thinking and may expose reliance on AI-generated answers.

Communication

Students may communicate with the instructor by email, during office hours, in person, or through scheduled virtual meetings, depending on the delivery format. I aim to respond to all inquiries within 24 hours, except on weekends, when responses may take longer. Course Announcements will be posted on Canvas, and you will receive email notifications through the university email system. If your question may also benefit other students and is not personal in nature, please use the FAQ Board on Canvas and post your question using threaded replies.

Course Materials and Resources

No textbook is required. Course materials will be posted throughout the semester on Canvas. I recommend that you purchase (or borrow) one of the following econometrics textbooks, which will be a helpful reference both now and in the future.

- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd Edition. Cambridge, MA: MIT Press.
- Greene, W. H. (2018). *Econometric Analysis*, 8th Edition. Pearson.

Other readings will be assigned from various news articles, journals, books, and research bulletins through Canvas.

Suggested reference texts:

- (Undergraduate-level econometrics) Wooldridge, J. M. (2016). *Introductory Econometrics: A Modern Approach*, 6th Edition. Cincinnati, OH: South-Western College Publishing.
- (Causal inference) Angrist, J. D., & Pischke, J.S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton: Princeton University Press.
- (Industrial Organization) Carlton, D. W. & J. M. Perloff (2005). *Modern Industrial Organization*, 4th Edition. Boston, MA: Pearson Addison Wesley.

Learning Objectives

By the end of the course, students will be able to:

- Derive an econometric estimating equation from a theoretical model of economic behavior
- Mathematically manipulate reduced-form regression coefficients to recover structural parameters for an underlying economic model
- Articulate the distinction between model calibration and causal inference and identify the conditions necessary to achieve the latter
- Efficiently and effectively manipulate data and run regressions in statistical programming software

- Correctly report and interpret regression results
- Apply various identification strategies to data in realistic research settings
- Describe and implement the theoretical and empirical models presented in the lecture

Delivery Format

This course may be offered face-to-face or online. Both delivery formats cover the same course content, address the same student learning outcomes, and maintain equivalent academic requirements and expectations. Differences between the formats relate primarily to the mode of instruction, communication, and student interaction. Online students access course materials and instructional activities through Canvas and communicate with the instructor and other students electronically and/or through scheduled virtual meetings.

Course Requirements and Grading

Evaluation of Performance: Students can earn a total of 500 possible points in this course. The final course grade will have the following benchmarks out of the 500 possible grade points.

Exams (3 at 75 points each)	225 points
Homework Assignments (4 at 50 points each)	200 points
Quizzes (3 at 15 points each)	45 points
Class Participation-TA (3 at 10 points each)	30 points
Total	500 points

Grades in this course will be assigned based on absolute performance without applying a curve. Letter grades will be assigned according to the following scheme.

Grading scale: A = 90-100%; B = 80-89%; C = 70-79%; D = 60-69%; F = 59% and below

Exams: There are three closed-note, in-person exams. Regardless of the course delivery format (Campus 1 or Campus 5), all students are required to take the exams in person. The exams will be administered on the Mississippi State University campus at the scheduled time and location.

Exam 1: **RM 103 LRW in March (in-class)**

Exam 2: **RM 103 LRW in April (in-class)**

Exam 3: **RM 103 LRW, TBA, during the final exam week (May 5-11, 2027)**

Exams will require students to solve theory-based problems and interpret results excerpted from the empirical literature.

Make-up exams will not be given except in the case of prior approval from the instructor. If you must miss an exam, the absence must be for a University-recognized reason (authorized university activity or other reasons as stated by the University's Academic Operating Policy & Procedure document, [AOP 12.09](#)) and **MUST** adhere to the following procedure:

- You must contact the instructor before the exam. Email me at a.kim@msstate.edu AND follow up appropriately via email or online meeting.
- Official documentation must be provided as proof of absence.

There will be no exceptions to these requirements. Provided both are met, the request for a make-up exam will be considered and may be permitted at my sole discretion. If both requirements are not met, a grade of zero will be recorded. This is non-negotiable.

Reading quizzes: may be administered during scheduled class meetings or through Canvas, depending on the delivery format. The quizzes will be short and include relatively straightforward questions designed to assess whether students have actually completed the reading. The quizzes will be open-text and open-notes; students need only be familiar with the reading, and do not need to memorize material.

***Additional readings, reading quizzes, and/or short assignments may be assigned on an ad-hoc basis at the instructor's discretion. All such assignments will contribute to the "reading quizzes" component of a student's final grade.

Homework assignment: will be a combination of theory-based problems and empirical exercises using R. Assignments will be made available to students at least a week before their due date. Without prior approval by the instructor, late submission will be reduced by 20%. Assignments submitted more than one week late will not receive credit. Collaboration is allowed/encouraged on assignments; however, each student must submit her/his own work.

Class Participation: includes active engagement in class meetings and/or online instructional activities, depending on the delivery format. This may involve student-led activities such as presenting solutions to assigned problems, explaining derivations, or presenting and discussing code developed during lab sessions. Students are expected to contribute to discussions by asking questions, offering insights, and engaging constructively with their classmates' presentations.

Academic Misconduct: Academic honesty is expected. This means that cheating and plagiarism will **NOT** be tolerated. If discovered, you will receive a zero for the assignment, the quiz, and the exam. Continued disregard for this policy could result in an "F" grade for the course. All students are expected to be familiar with the Student Honor Code (see Policies on page 6).

Brief Topical Outline

Subject		Approx. # Classes
I	Introduction and Scope	4
II	Consumer Demand	9
III	Agricultural Supply	6
IV	Supply and Demand	6
V	Market Structure and Imperfect Competition	2
VI	In-class Exams	2
Total		29

Tentative Course Outline

Topic
1. Course Overview Course preliminaries and introduction
2. Utility Maximization and Consumer Demand a. Cobb-Douglas model of consumer demand & wage b. Linear expenditure system c. The “Almost Ideal Demand System” and its extensions and applications d. Stone’s price index and the linear approximate AIDS model e. Testing economic theory using the LA/AIDS model
3. Agricultural Supply - Modeling Price Expectations a. Adaptive expectations supply model b. Partial adjustment supply model c. Nerlovian agricultural supply model
4. Simultaneous Estimation of Supply and Demand a. Instrumental variables and their application to the problem of endogeneity b. Indirect Least Squares c. Two-Stage Least Squares d. Simultaneous Estimation and modern applications
5. Market Structure and Imperfect Competition a. Modeling market power and Cournot duopolies b. Empirical methods for estimating market power c. Monopsony, oligopsony, and estimating buying power

Key Dates

January 13	Class Begin
January 20	Last day to drop a course without a grade (11:59 p.m.)
January 21	Last day to add a course (5:00 p.m.)
March 4	Last day to drop a course with "W" grade (5:00 p.m.)
March 8 to March 14	Spring Break - No class
May 4	Last class day

Policies

Generative AI

Mississippi State University recognizes the introduction of generative artificial intelligence (generative AI) provides opportunities for scholarly rigor, intellectual integrity, and educational excellence. Nevertheless, the use of generative AI also poses issues related to academic integrity. Currently, individual instructors are encouraged to establish class-specific guidelines concerning the use of generative AI within their courses. The student must consult the syllabus for each class they are taking to determine if and to what degree generative AI is allowed. In this course, assessments are designed to evaluate students’ analytical reasoning, model setup, and identification strategies.

Disability Resource Center

Mississippi State University is committed to providing equitable access to learning opportunities for all students. The Disability Resource Center (01 Montgomery Hall) collaborates with students who have disabilities to arrange reasonable accommodations. If you have, or think you may have, a disability, please contact drc@saffairs.msstate.edu or 662-325-3335 to arrange a confidential discussion regarding equitable access and reasonable accommodations. Disabilities may include, but are not limited to, conditions related to mental health, chronic health, attention, learning, autism, brain injury, vision, hearing, mobility, speech, or intellectual disabilities. In the case of short-term disabilities (e.g., broken arm), students and instructors can often work to minimize barriers. If additional assistance is needed, please contact the Disability Resource Center. For more information, visit <https://www.drc.msstate.edu/>.

Student Honor Code

Mississippi State has an approved Honor Code that applies to all students. The code is as follows: "As a Mississippi State University student, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do." Upon accepting admission to Mississippi State University, a student immediately assumes a commitment to uphold the Honor Code, to accept responsibility for learning, and to follow the philosophy and rules of the Honor Code. Student will be required to state their commitment on examinations, research papers, and other academic work. Ignorance of the rules does not exclude any member of the MSU community from the requirements or the processes of the Honor Code. For additional information, please visit: <http://honorcode.msstate.edu/policy>

Title IX

MSU is committed to complying with Title IX, a federal law that prohibits discrimination, including violence and harassment, based on sex. This means that MSU's educational programs and activities must be free from sex discrimination, sexual harassment, and other forms of sexual misconduct. If you or someone you know has experienced sex discrimination, sexual violence and/or harassment by any member of the University community, you are encouraged to report the conduct to MSU's Direction of Title IX/EEO Programs at 325-8124 or by email to titleix@msstate.edu. Additional resources are available at [Dean of Students - Sexual Misconduct and Sexual Assault](#).

Mandatory Reporting

Faculty members at Mississippi State University are required to report any information received about possible sexual misconduct. This includes information shared in class discussions or assignments, as well as information shared in conversations outside class. The purpose of reporting is to allow MSU to take steps to ensure a safe learning environment for all. The university also has confidential resources available, who can provide assistance to those who have experienced sexual misconduct without requiring a mandatory reporting duty. Students may access confidential campus resources here: [Office of Civil Rights Compliance - Title IX and Sexual Misconduct](#).

University Safety Statement

Mississippi State University values the safety of all campus community members. Students are encouraged to register for Maroon Alert texts and to download the Everbridge App. Visit the Personal Information section in Banner on your MyState portal to register. To report suspicious activity or to request a courtesy escort via Safe Walk, call University Police at [662-325-2121](tel:662-325-2121), or in case emergency, call [911](tel:911). For more information regarding safety and to view available training including helpful videos, visit ready.msstate.edu.